De Mathematics Competitions

4th Annual

DMC 10 S

Sunday, September 27, 2026



INSTRUCTIONS

1. DO NOT TURN TO THE NEXT PAGE UNTIL YOUR COMPETITION MANAGER TELLS YOU TO BEGIN.
2. This is a 25-question multiple-choice competition. For each question, only one answer choice is correct.
3. Mark your answer to each problem on the answer sheet with a #2 pencil. Check blackened answers for accuracy and erase errors completely. Only answers that are properly marked on the answer sheet will be scored.
4. SCORING: You will receive 6 points for each correct answer, 1.5 points for each problem left unanswered, and 0 points for each incorrect answer.
5. Only blank scratch paper, rulers, compasses, and erasers are allowed as aids. No calculators, smartwatches, phones, or computing devices are allowed. No problems on the competition will require the use of a calculator.
6. Figures are not necessarily drawn to scale.
7. Before beginning the competition, your competition manager will ask you to record your name on the answer sheet.
8. You will have 75 minutes to complete the competition once your competition manager tells you to begin.
9. When you finish with the competition, sign your name in the space provided on the answer sheet and complete the demographic information questions on the back of the answer sheet.

The problems and solutions for this DMC 10 were prepared
by the DMC Committee under the direction of:

Taiki Aiba and Ryan Shin

The DMC Committee reserves the right to disqualify scores from a school if it determines that the rules or the required security procedures were not followed.

The publication, reproduction or communication of the problems or solutions of this competition during the period when students are eligible to participate seriously jeopardizes the integrity of the results. Dissemination via phone, email, or digital media of any type during this period is a violation of the competition rules.

Students who score well on this DMC 10 will be invited to take the De Invitational Mathematics Examination (DIME), which is already out and is due on Friday, January 29, 2027. More details about the DIME can be found on our site.

1. A red container is filled with water, and 20% of its water is poured into an empty blue container. What is the ratio of the amount of water in the blue container to the amount of water in the red container?

   (A) 1 : 6 **(B)** 1 : 5 **(C)** 1 : 4 **(D)** 1 : 3 **(E)** 1 : 2

2. Katie and Jim are standing in a row of people, with Katie somewhere to the right of Jim. There are 7 people to the left of Katie, 5 people to the right of Jim, and 2 people between Katie and Jim. How many people are in the row?

   (A) 10 **(B)** 11 **(C)** 12 **(D)** 13 **(E)** 14

3. What is the least positive integer n such that $n! + 1$ is not divisible by any integer from 2 to 9, inclusive?

   (A) 4 **(B)** 5 **(C)** 6 **(D)** 7 **(E)** 8

4. A right triangle has legs of lengths 5 and 6. A square is drawn with the hypotenuse of the triangle as one of its sides, so that the triangle lies inside the square. What is the area of the region inside the square but outside the triangle?

   (A) 46 **(B)** 47 **(C)** 48 **(D)** 49 **(E)** 50

5. Ken writes 10 positive integers, not necessarily distinct, on a sheet of paper. John then asks him the following questions:

   - How many of the numbers on your paper are less than 2?
   - How many of the numbers on your paper are greater than 2?
   - How many of the numbers on your paper are equal to 4?

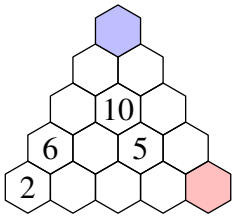
   Ken answers each question truthfully, and each answer is 3. What is the sum of the numbers on Ken's paper?

   (A) 21 **(B)** 22 **(C)** 23 **(D)** 24 **(E)** 25

6. Justin has three empty boxes of negligible weight and four pebbles, each weighing 3, 4, or 5 ounces. He places each pebble in one of the boxes so that every box contains at least one pebble and the weights of the three boxes are distinct and form an arithmetic sequence. What is the greatest possible weight, in ounces, of the heaviest box?

   (A) 4 **(B)** 5 **(C)** 6 **(D)** 7 **(E)** 8

7. In the figure below, the number in each hexagon above the bottom row is equal to the sum of the numbers in the two hexagons directly below it. How much greater is the number in the blue hexagon than the number in the red hexagon?



(A) 39 **(B)** 40 **(C)** 41 **(D)** 42 **(E)** 43

8. A rectangle with perimeter 40 is divided into four smaller rectangles by one line parallel to each pair of its sides. Three of the smaller rectangles have perimeters 19, 21, and 25. What is the perimeter of the fourth?

(A) 9 **(B)** 15 **(C)** 17 **(D)** 23 **(E)** 27

9. Arn, Bob, Cyd, Dan, and Eve are each given a different one of the numbers 3, 12, 17, 23, and 34. Arn does not have the greatest number, the two numbers closest to each other belong to Cyd and Dan, and Eve's number is odd. Which number does Bob have?

(A) 3 **(B)** 12 **(C)** 17 **(D)** 23 **(E)** 34

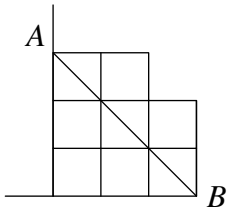
10. Daniel walks one mile in four straight segments, each a quarter mile long. After each of the first three segments, he turns 90° either clockwise or counterclockwise, each with probability $\frac{1}{2}$. What is the probability that he ends where he started?

(A) $\frac{1}{8}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{2}$ **(D)** $\frac{5}{8}$ **(E)** $\frac{3}{4}$

11. The numbers a, b, and c are three consecutive positive integers in some order, and so are p, q, and r. Also, $a \cdot p = 161$ and $b \cdot q = 189$. What is $c \cdot r$?

(A) 128 **(B)** 144 **(C)** 160 **(D)** 176 **(E)** 192

12. Peter places a row of $n \geq 3$ unit squares on the floor, pushed against a wall. He then places a row of 3 unit squares on top of it and a row of 2 unit squares on top of that, each row also pushed against the wall. The figure below shows the resulting shape for $n = 3$. Let A and B be the top-left and bottom-right vertices of the shape, respectively. What is the greatest value of n for which segment $\overline{AB}$ does not go outside the shape?



(A) 3 **(B)** 4 **(C)** 5 **(D)** 6 **(E)** 7

13. In a survey, each respondent is either a truth-teller, who always tells the truth, or a liar, who always lies. Each respondent's favorite number is either 2 or 5. Every respondent answered two questions:

    - Is your favorite number 2?
    - Is your favorite number either 2 or 5?

    Exactly 38 liars have favorite number 2, 87 respondents answered "Yes" to the first question, and 106 respondents answered "No" to the second question. How many truth-tellers have favorite number 2?

    (A) 16 **(B)** 17 **(C)** 18 **(D)** 19 **(E)** 20

14. Kevin chooses two different positive integers and writes their sum, their positive difference, and their product on a board. Exactly one of the three numbers on the board is even, and it is 72. What is the sum of all possible values of the smaller of Kevin's two integers?

    (A) 5 **(B)** 6 **(C)** 8 **(D)** 11 **(E)** 12

15. In rectangle $ABCD$ with $AB = 5$ and $BC = 4$, point E lies on side $\overline{CD}$ so that segment $\overline{AE}$ bisects $\angle BED$. What is AE?

    (A) $3\sqrt{2}$ **(B)** $2\sqrt{5}$ **(C)** $3\sqrt{3}$ **(D)** $2\sqrt{7}$ **(E)** $\sqrt{30}$

16. Aki and Hanami each have a positive integer from 1 to 20, inclusive. Each knows only their own number and that at least one of the two numbers is odd. Aki says, "My number is greater than 4 but less than 7." Hanami says, "Now I know that my number is at least three times yours." Aki says, "Now I know your number." What is the sum of Aki's and Hanami's numbers?

    (A) 21 **(B)** 22 **(C)** 23 **(D)** 24 **(E)** 25

17. Yui picks 5 integers from 1 to 35, inclusive, so that any two of them differ by at least 4. Three of her numbers are 16, 21, and 30. How many possibilities are there for the other two numbers?

    (A) 86 **(B)** 87 **(C)** 88 **(D)** 89 **(E)** 90

18. In a classroom, each student has exactly two pencils, and each pencil is orange, yellow, or gray. Exactly 56 students have at least one orange pencil, 48 students have no gray pencil, and 34 students have two pencils of different colors. How many students have at least one yellow pencil?

    (A) 26 **(B)** 27 **(C)** 28 **(D)** 29 **(E)** 30

19. Parallelogram $ABCD$ has $BC = 5$, $\angle ABC < 90°$, and $\angle ACB > 90°$. The circle through A, B, and C meets line AD again at P and side $\overline{CD}$ again at Q. Also, $CP = 10$ and $CQ = 4$. What is AP?

    (A) $\frac{48}{7}$ **(B)** 7 **(C)** $\frac{36}{5}$ **(D)** $\frac{15}{2}$ **(E)** 8

20. A sequence $a_1, a_2, \ldots, a_6$ of positive integers has the property that every term is equal to at least one other term of the sequence. How many different values less than 1000 can the product $a_1 a_2 \cdots a_6$ take?

    (A) 36 **(B)** 37 **(C)** 38 **(D)** 39 **(E)** 40

21. In equilateral $\triangle ABC$, points D and E lie on lines AB and AC, respectively, both on the opposite side of line BC from A. Also, $CE = DE$, and the circumcircle of $\triangle CDE$ is tangent to line AB at D. What is the degree measure of $\angle CDE$?

    (A) 70 **(B)** 72 **(C)** 75 **(D)** 80 **(E)** 84

22. Isosceles trapezoid $ABCD$ with $\overline{AB} \parallel \overline{CD}$, $AB = AD = BC = 3$, and $CD = 7$ is inscribed in a circle. Point E on the circle satisfies $\overline{BE} \perp \overline{AC}$. What is DE^2?

    (A) 24 **(B)** 26 **(C)** 28 **(D)** 30 **(E)** 32

23. Joy picks an integer n from 1 to 40, inclusive. She tells Amy the remainder when n is divided by 7, and she tells Sid the number of positive divisors of n. Amy and Sid both know that $1 \leq n \leq 40$, but they mistakenly believe that Amy was told the number of divisors and Sid was told the remainder. Amy says, "I know what n is." Sid replies, "Then I also know what n is." Both statements are true according to their beliefs, and Amy and Sid have deduced the same number, but that number is not n. What is the sum of all possible values of n?

    (A) 21 **(B)** 22 **(C)** 23 **(D)** 24 **(E)** 25

24. What is the sum of the digits of the least positive integer n such that

$$\sqrt{5n-1} - \sqrt{5n-2} + \sqrt{5n-3} - \sqrt{5n-4}$$

is less than 0.05?

(A) 8 **(B)** 9 **(C)** 10 **(D)** 11 **(E)** 12

25. In right $\triangle ABC$ with a right angle at A and $BC = 7$, let D be the midpoint of side $\overline{AB}$. There is a point E on the circumcircle of $\triangle ABC$ such that $\triangle CDE$ is equilateral. What is the side length of $\triangle CDE$?

(A) $\sqrt{21}$ **(B)** $2\sqrt{7}$ **(C)** $\frac{7\sqrt{21}}{6}$ **(D)** $\frac{7\sqrt{10}}{4}$ **(E)** $3\sqrt{6}$